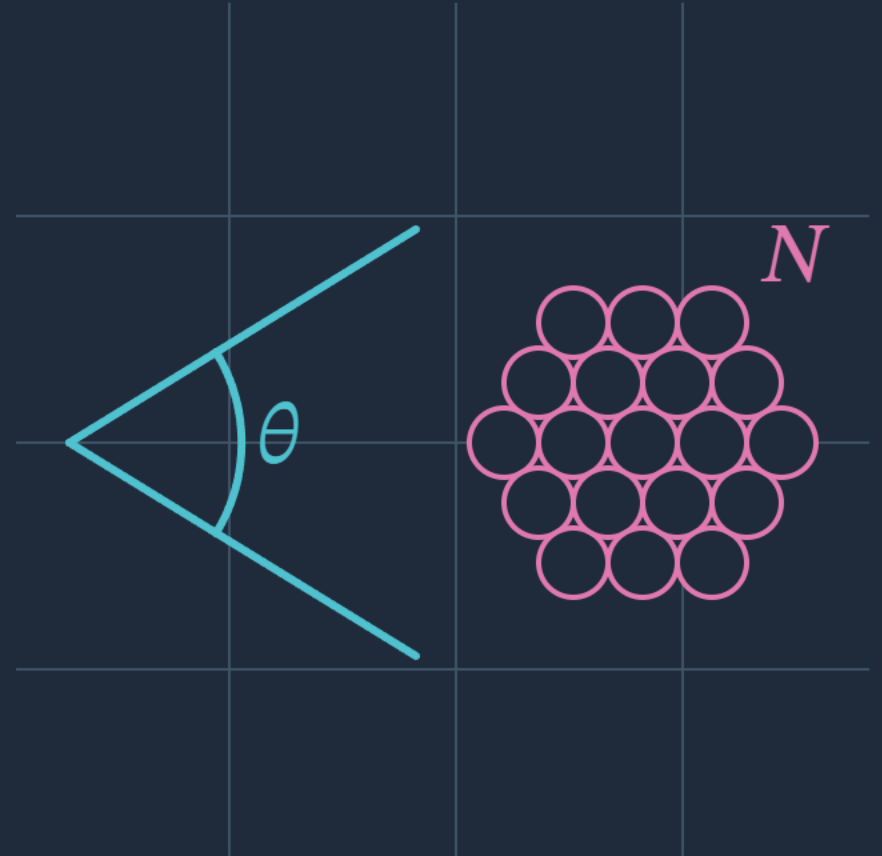


When Should a Quantity Have Its Own Dimension?

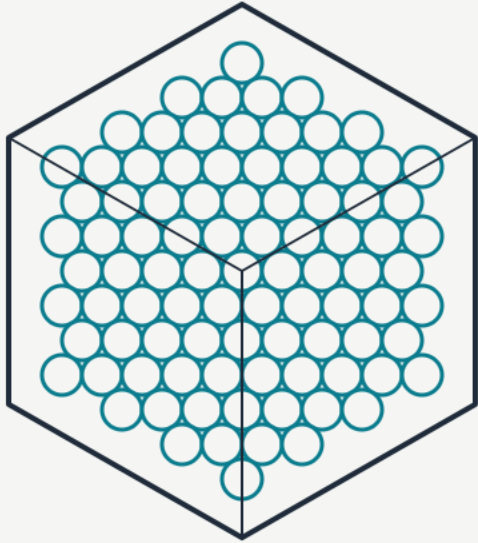
Uniqueness of natural units as a design criterion

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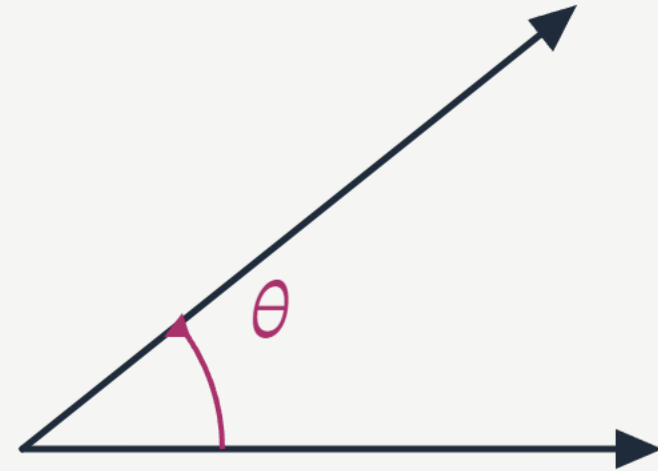


An asymmetric choice in the SI



Amount of substance (counting)

→ Has a dimension (mol)

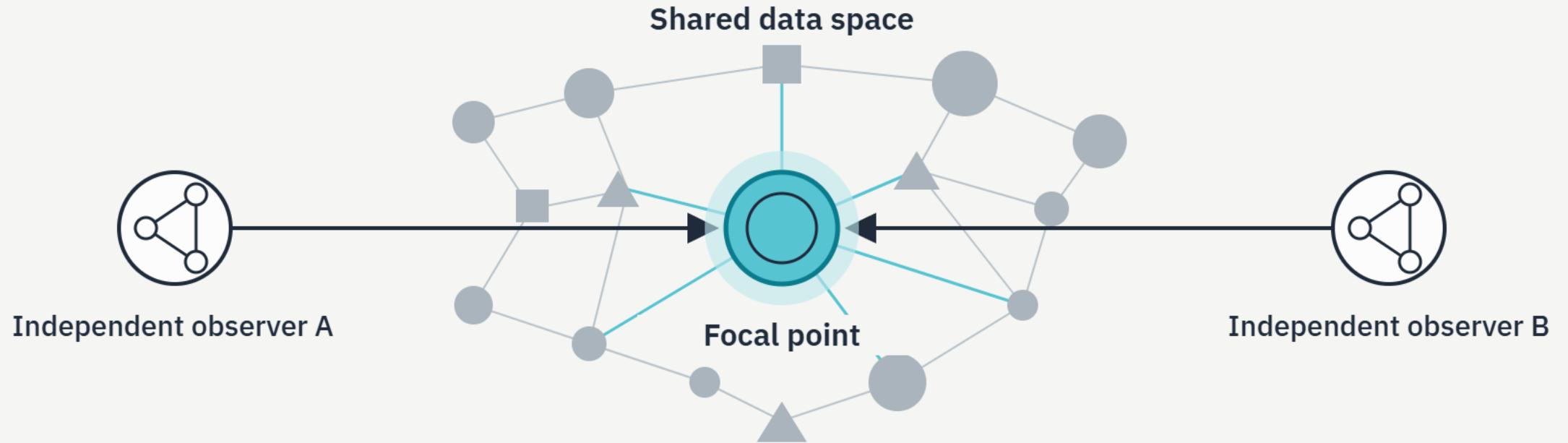


Plane angle

→ Dimensionless ($1 \text{ m/m} = 1$)

Why does one have a dimension and the other not?
A universal criterion to adjudicate between the two is missing.

A new criterion: uniqueness of the natural unit

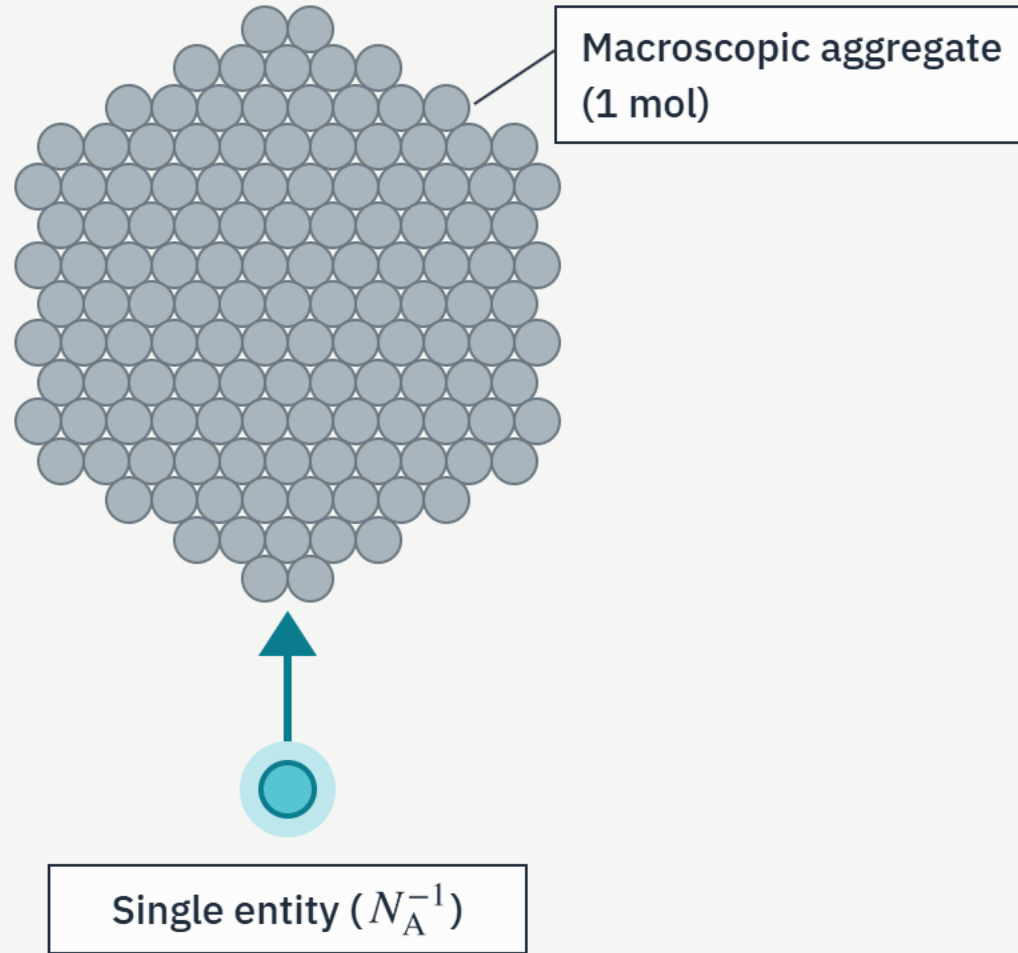


A quantity can be communicated by “pure numbers” alone, without loss of information, only when independent parties can converge on the same reference without prior communication — that is, only when there is exactly one salient natural unit.

Diagnostic matrix: the structure that decides whether a dimension is needed

Kind of quantity	Candidate focal points	Uniqueness	Verdict
Amount of substance	Single entity (N_A^{-1})	☑ Unique	Dimension optional (the mole is legitimate)
Plane angle	Radian vs full circular angle (Ω_1)	✗ Not unique	Own dimension required
Solid angle	Steradian vs full spherical solid angle (Ω_2)	✗ Not unique	Own dimension required
Logarithmic quantities	$\log e$, $\log 2$, $\log 10$	✗ Not unique	Own dimension required

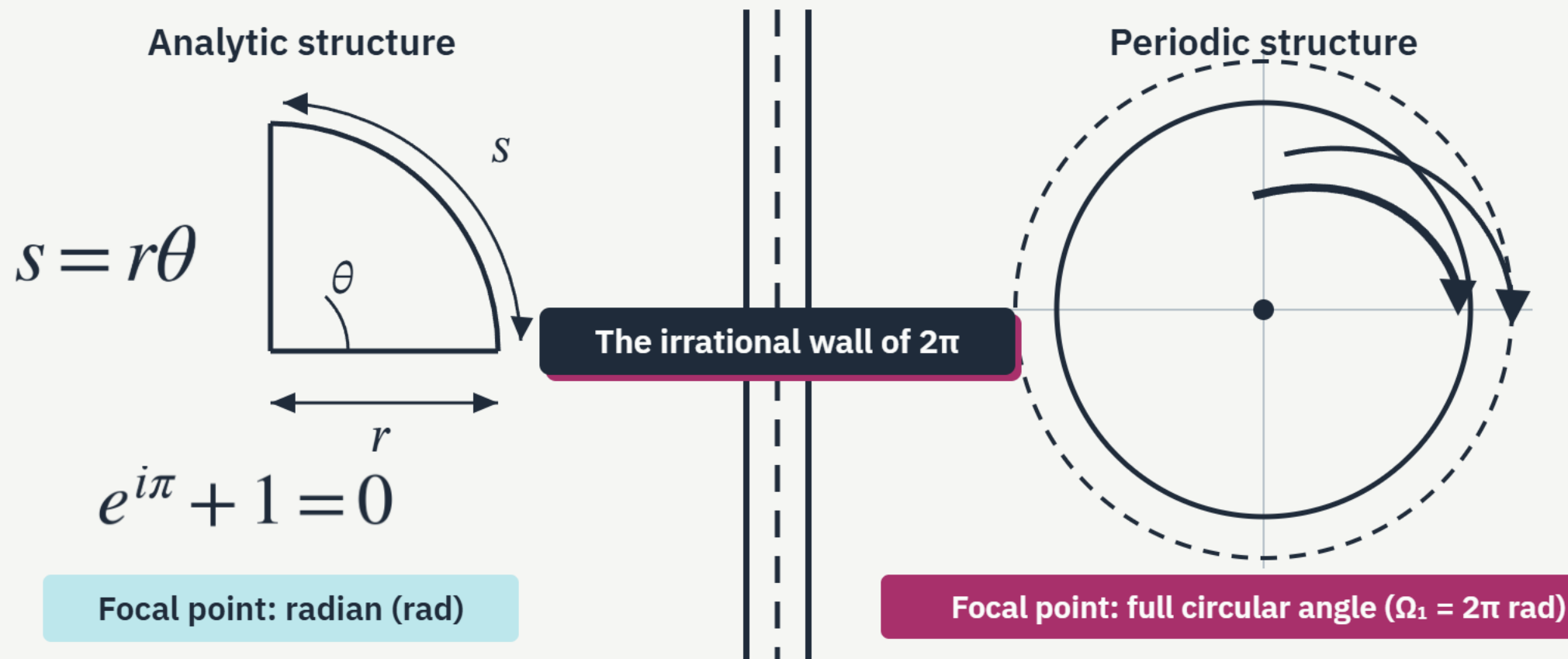
Test 1: amount of substance (a unique case)



The amount of substance has no natural unit other than the “single entity.” Because the focal point is unique, nothing is lost by expressing it as a pure number.

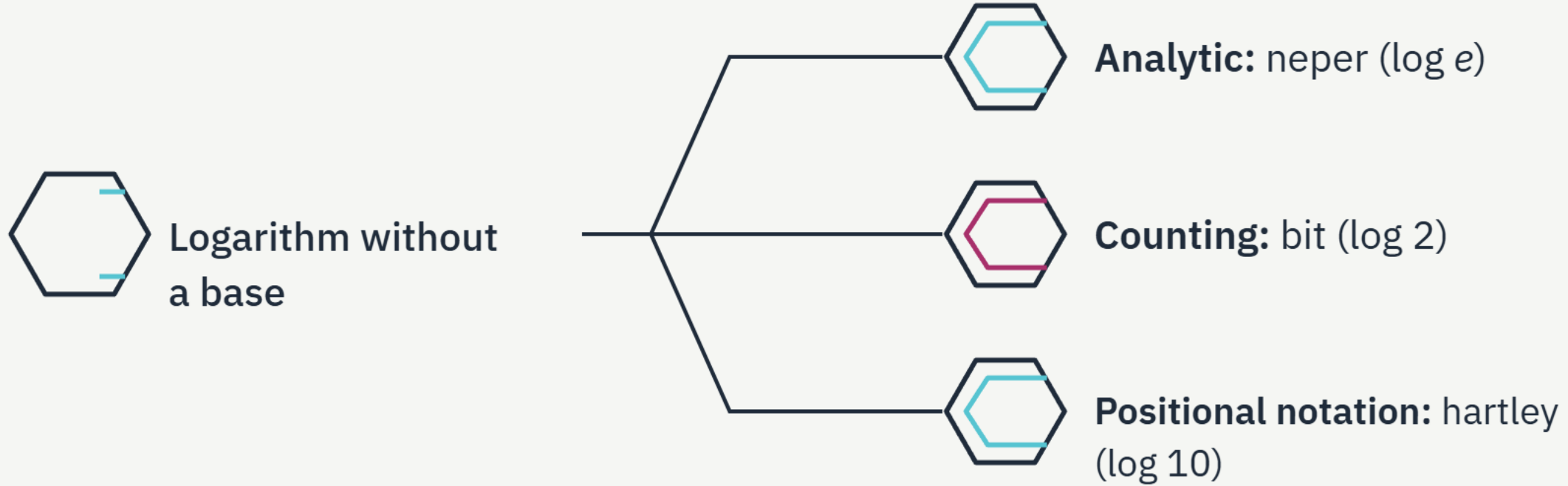
The dimension is optional, and the SI’s choice to give it the mole is also a legitimate one.

Test 2: plane angle (a non-unique case)



The angle has two powerful natural units, each salient for an entirely different reason. They are incommensurable, so neither can be made the sole reference. An independent dimension is therefore required.

Test 3: logarithmic quantities (a non-physical case)



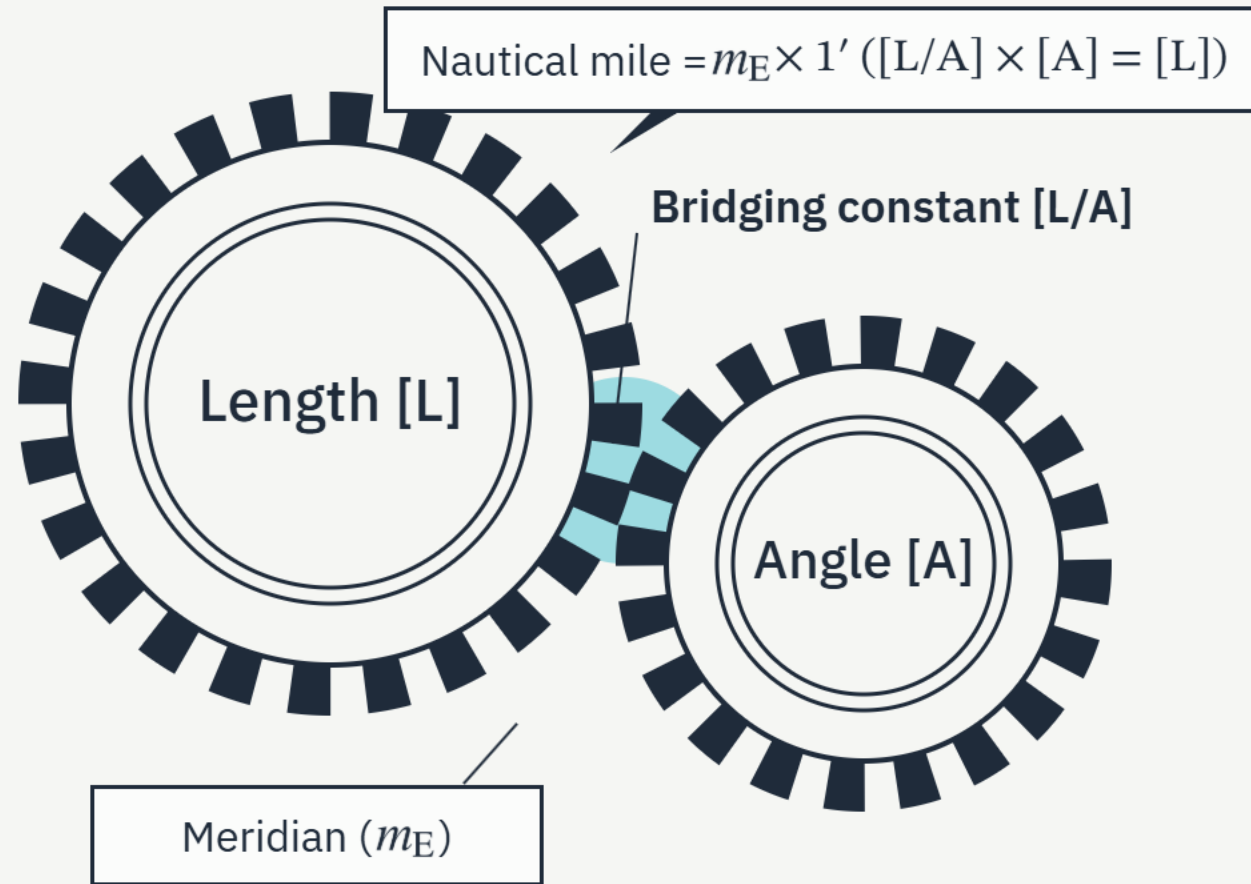
The same phenomenon occurs even for mathematical objects in which no physical law is involved. Since three natural focal points stand side by side, logarithmic quantities should be treated as a dimension in information theory as well.

“Loss of information” through de-dimensionalization



When a non-unique quantity is made dimensionless, which unit was replaced by “the number 1” becomes a black box. Dropping the dimension amounts to hiding the convention on which it rests.

The solution: an elegant category called “bridging constants”

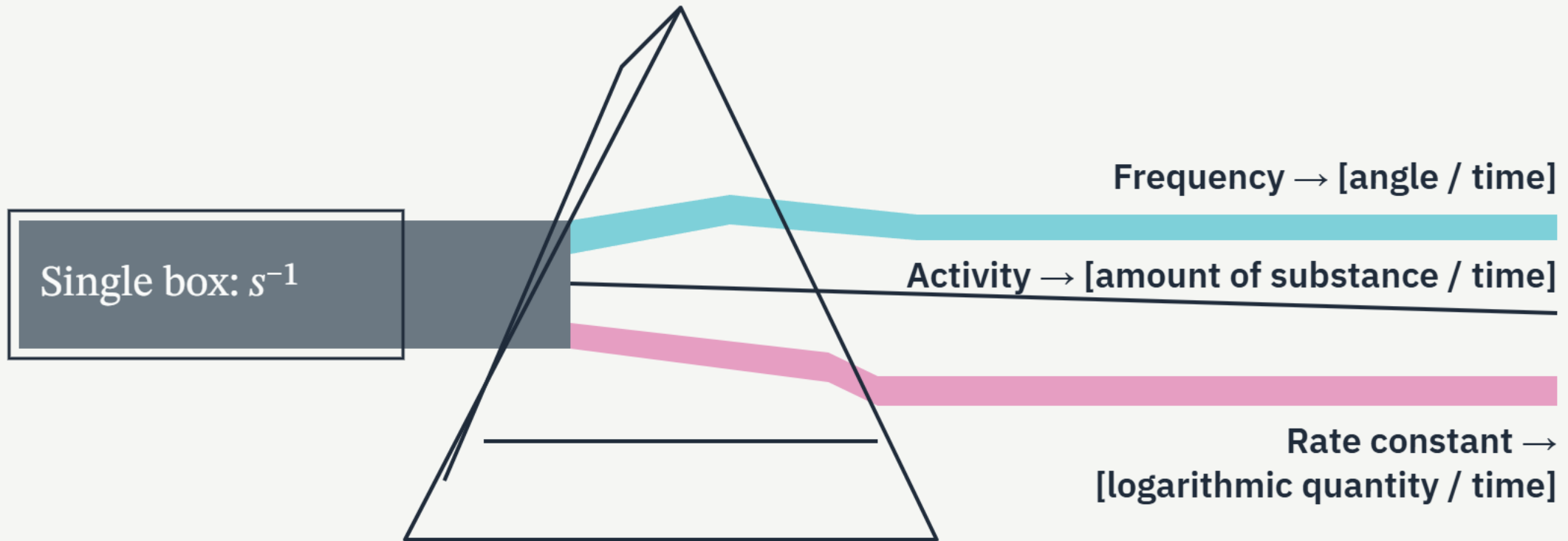


Multiplying the minute, of dimension [angle], by the bridging constant called the meridian, of dimension [length/angle], gives the nautical mile, of dimension [length].

The meridian belongs to a family of constants with a clear meaning: they convert a periodic variable into a linear one.

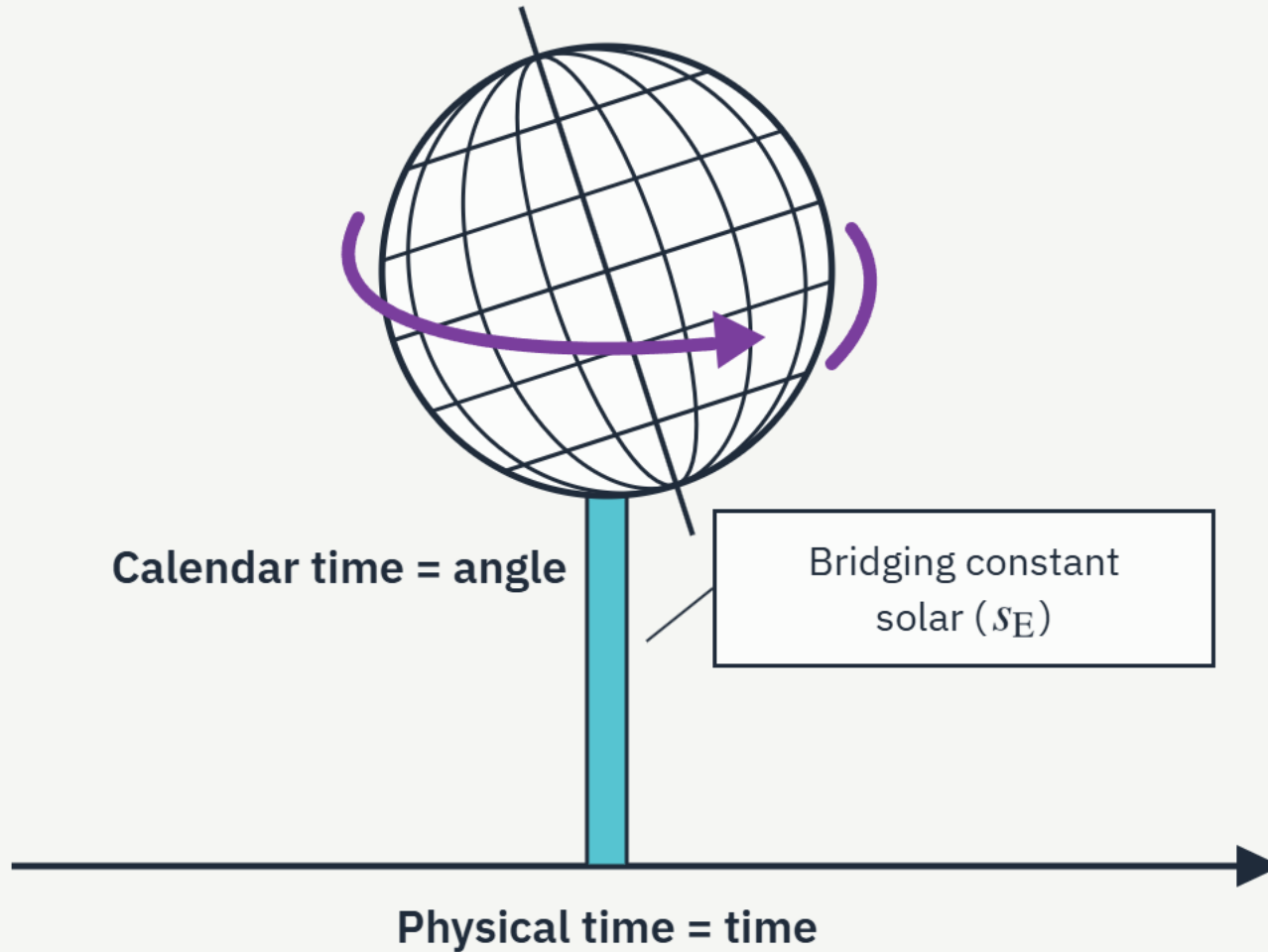
Bridging constants $\ni m_E$

Benefit 1: a clean resolution of s^{-1}



Three different concepts that the SI packs into a single box separate naturally under dimensional analysis. What “one event” means in each phenomenon is stated plainly by the dimension of the equation itself.

Benefit 2: separating calendar time from physical time



“Calendar time,” based on the Earth’s rotation, and uniform “physical time” are by nature of different dimensions. The two are linked by a bridging constant.

The leap-second problem is that the bridging constant called the solar actually varies.

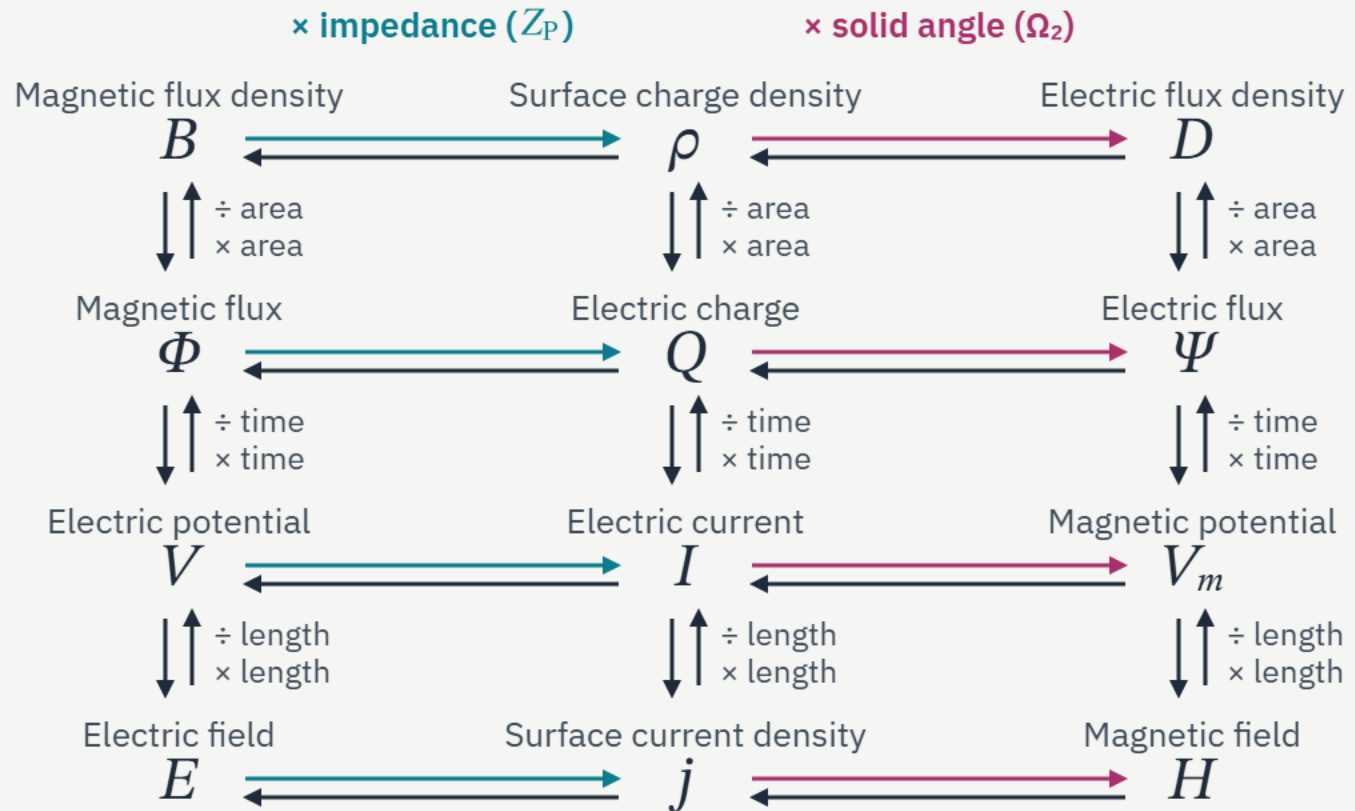
Bridging constants $\ni s_E$

Benefit 3: symmetry of electromagnetism in five base dimensions

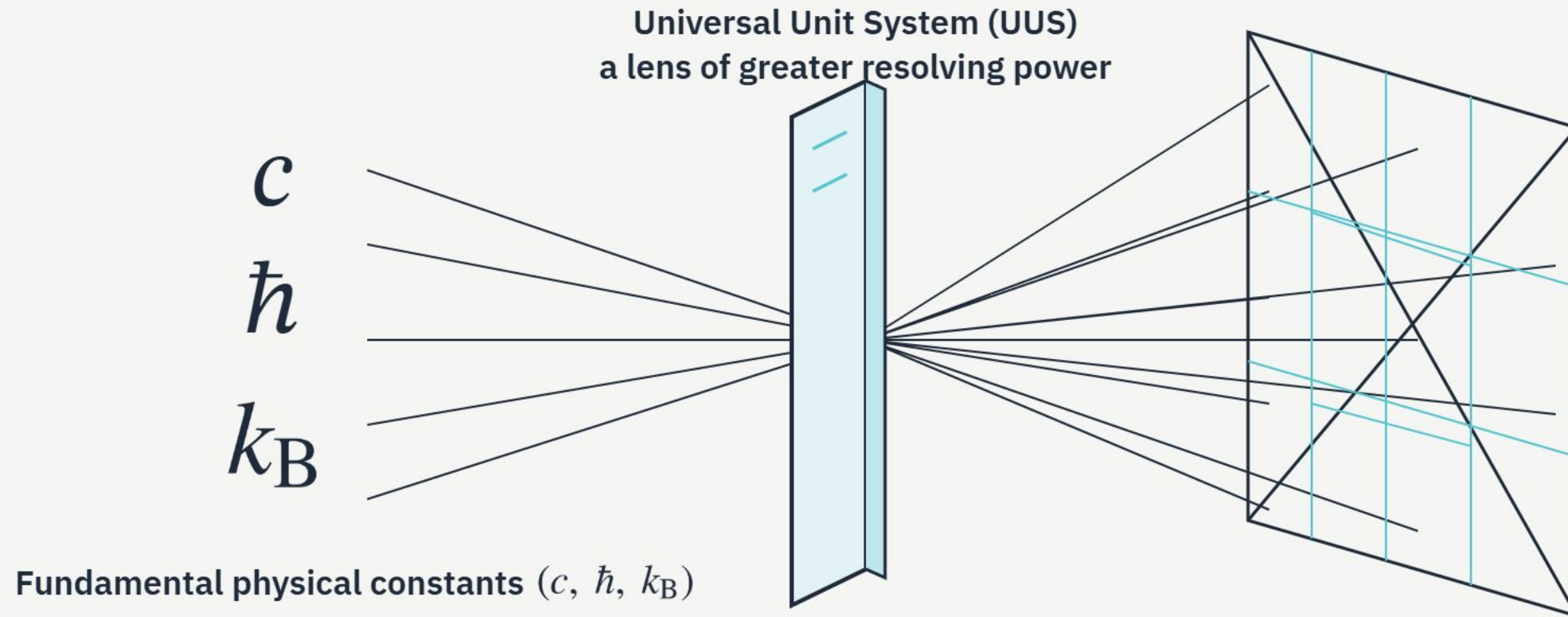
MKSZ_PΩ₂

Giving the solid angle its own dimension reveals a striking symmetry.

Field strengths, potentials, fluxes and densities fall into a perfectly symmetric electric–magnetic grid.

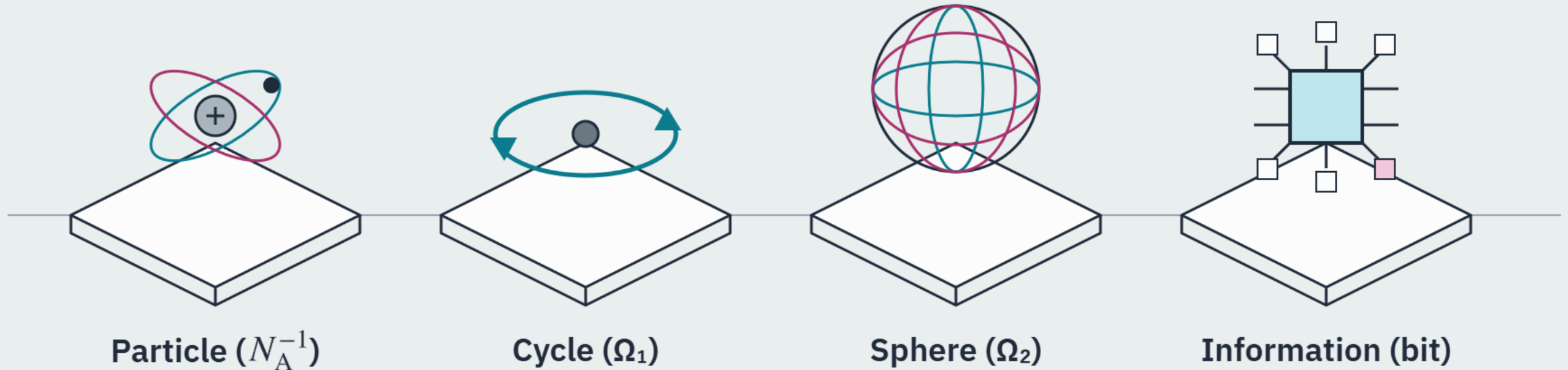


The Universal Unit System (UUS) as an existence proof



This theory is not mere speculation: it is fully implemented as a working system of units.
It does not reject the SI; it is a framework of greater resolving power that offers a complementary viewpoint.

Conclusion: dimensions as semantic markers



Dimensions are not constraints that cramp calculation.

They are semantic markers through which an equation tells, by itself, what it is.

By restoring the natural units that were not chosen, physical laws show a deeper clarity.